

A SPICE Circuit Optimiser (ASCO)

An open source optimiser for the existing SPICE simulators, ASCO has its codes available under the GNU GPL licence. Designed to address the problems related to electric circuits, ASCO is the result of interaction with experienced designers. Try out ASCO included in this month's DVD, available with EFY Plus

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A SPICE circuit optimiser, popularly known as ASCO, has found its way through an academic research project that intended to design low-voltage, low-power and high-performance analogue circuits for mobile communications. SPICE (simulation program with integrated circuit emphasis) is a general-purpose, open source analogue electronic circuit simulator.

The main task of ASCO is neither to suggest nor to create new circuit arrangements but to simplify the design process. It automatically verifies, fine-tunes and optimises the functionality of the circuit for temperature and voltage throughout the process.

Features

ASCO has a set of in-built features due to which it is considered a great help for the designers in quest of designing a better circuit. Perhaps the most interesting is that, ASCO is simulator-independent. This means, it functions independent of the simulator it is working with. It thus provides out-of-the-box support for multiple simulators like ngspice, Qucs, LTspice, HSPICE, Spectre R and Eldo. That is not all, the future releases of ASCO are expected to include more simulators. Key features of the tool are described below.

(**Note.** If a simulator can be launched from the command line, can read the inputs from the text files, output the result in the ASCII format, it can be added to the list of the supported SPICE simulators. It flexibly works with tools like FastHenry.)

Number of variables. ASCO is hard-coded in 'C' and theoretically

Key functions of ASCO

- Fully redesigns a new circuit described in a SPICE netlist
- Initiates the circuit optimisation capabilities in existing SPICE simulators, using high-performance parallel differential evolution optimisation algorithm
- Reuses and optimises an existing circuit
- Simplifies the design process

Benefits of using ASCO

Works as an automation tool. ASCO does all the tedious tasks and gives time to a designer to concentrate on the solutions for the existing problems and the challenges related to architectures.

Easy migration. Designers can migrate a working and an existing design to an advanced semiconductor technology without much effort.

Explore new design space. It allows the designers to easily explore a new design space for an already existing topology so as to reduce the power consumption and area. ASCO allows the designers to find feasible topology even before deriving the equations to describe the circuit operation.

Increase in robustness and yield. ASCO increases the robustness and yield of a previously designed circuit just by guaranteeing that it complies with all the constraints in (all/some of) the process corners and all the design goals.

does not have a limit to the number of circuit variables that can be optimised. The only constraints that can be imposed are due to availability of the computer memory or the time required to generate a functional circuit.

Process-voltage-temperature (PVT) corners. Using the simulator functionality, ASCO can help determine the PVT corners for a designed circuit. The tests related to various design corners

and Monte Carlo analysis are limited only by capabilities of the simulator and to the time it takes to complete the optimisation process.

Efficiency. Using the differential evolution (DE) process, the optimisation algorithm features a global optimisation. Till date, it has been used in various applications producing great results in acceptable amount of time. As suggested in an open literature, this optimiser can also be looped in with a local optimiser to gain the convergence speed.

Parallel or distributed computation. There is simultaneous evaluation of the proposed values over multiple computers in one network so as to save time. ASCO does not have any limitation to the number of computers used or the number of values evaluated at a time.

No recompilation needed. Optimisation of arbitrary netlist in different conditions does not require recompilation of the code within the supported SPICE simulators.

File format. All the output data and the log information related to optimisation are stored in the plain text format so as to guarantee their readability in future. This format also allows the data and the files to be used by other existing tools after the optimisation process.

ASCO is a free software tool and its codes are available under the GNU GPL licence.

Optimisation algorithms

ASCO has both global and local optimisation algorithms, which makes this tool easily acceptable with many simulation tools. But what are those algorithms? We already know that ASCO uses DE for global optimisation, but is ASCO also capable of local optimisation?